FIG 1

Course map: from industrial minicomputer to measured microcontroller

A visual sequence of the curriculum spine



Learning arc

- Begin with why industrial computing exists: external events, consequences, and timing.
- Use the PDP-11 to teach architectural elegance and real-time software ideas.
- Use the MSP430 to make those ideas electrically observable on a modern bench.
- Use the Pico and Python to turn observations into evidence, not just impressions.

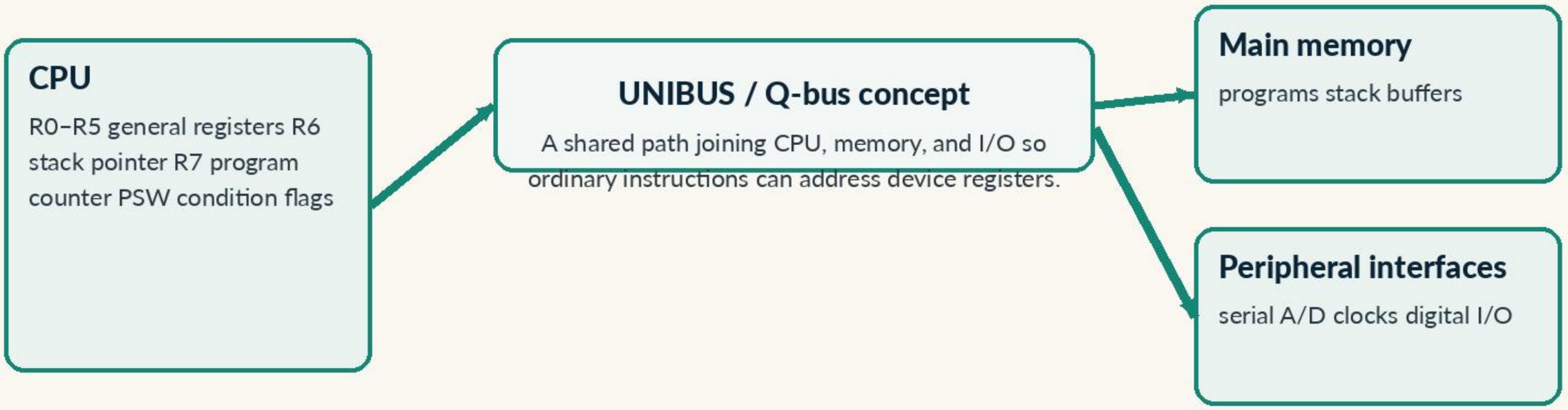
Curricular promise

- Every abstract concept must connect to a measurable consequence.
- Students should predict before they measure and publish what they actually observed.
- Historical context should explain design tradeoffs, not replace experimentation.

FIG 2

PDP-11 architecture in one teaching plate

Processor, bus, memory, and peripherals in the style needed for this curriculum



Why it mattered

Key teaching ideas

- Memory-mapped I/O: device registers live in the address space.
- Vectored interrupts: external devices can demand timely service.
- A modular bus let the same CPU serve laboratory, manufacturing, and communications roles.

Industrial interpretation

A controller is more than a CPU. It needs interfaces, deadlines, records, and a defined response to faults. The PDP-11 became important because it could be adapted to different plants and instruments without redesigning the whole computing core.

FIG 3

PDP-11 and MSP430 register model comparison

The family resemblance that motivates the course

PDP-11

Word size: 16 bits Registers: R0–R7 R6 = SP R7 = PC
Two-operand instructions Rich addressing symmetry
Condition codes: N Z V C

shared ideas

important tradeoffs

MSP430

Word size: 16 bits Registers: R0–R15 R0 = PC R1 = SP R2 = status / constant generator R3 = constant generator
Two-operand instructions Condition codes: N Z V C

Teaching emphasis

- The MSP430 feels familiar to a PDP-11 programmer because registers, flags, and two-operand instructions play similar roles.
- The differences are equally important: more registers, different encoding pressures, and less symmetry in destination addressing.
- Students should compare source code, assembler listings, and measured behavior rather than relying on verbal analogy alone.

FIG 4

Addressing modes as visible machine actions

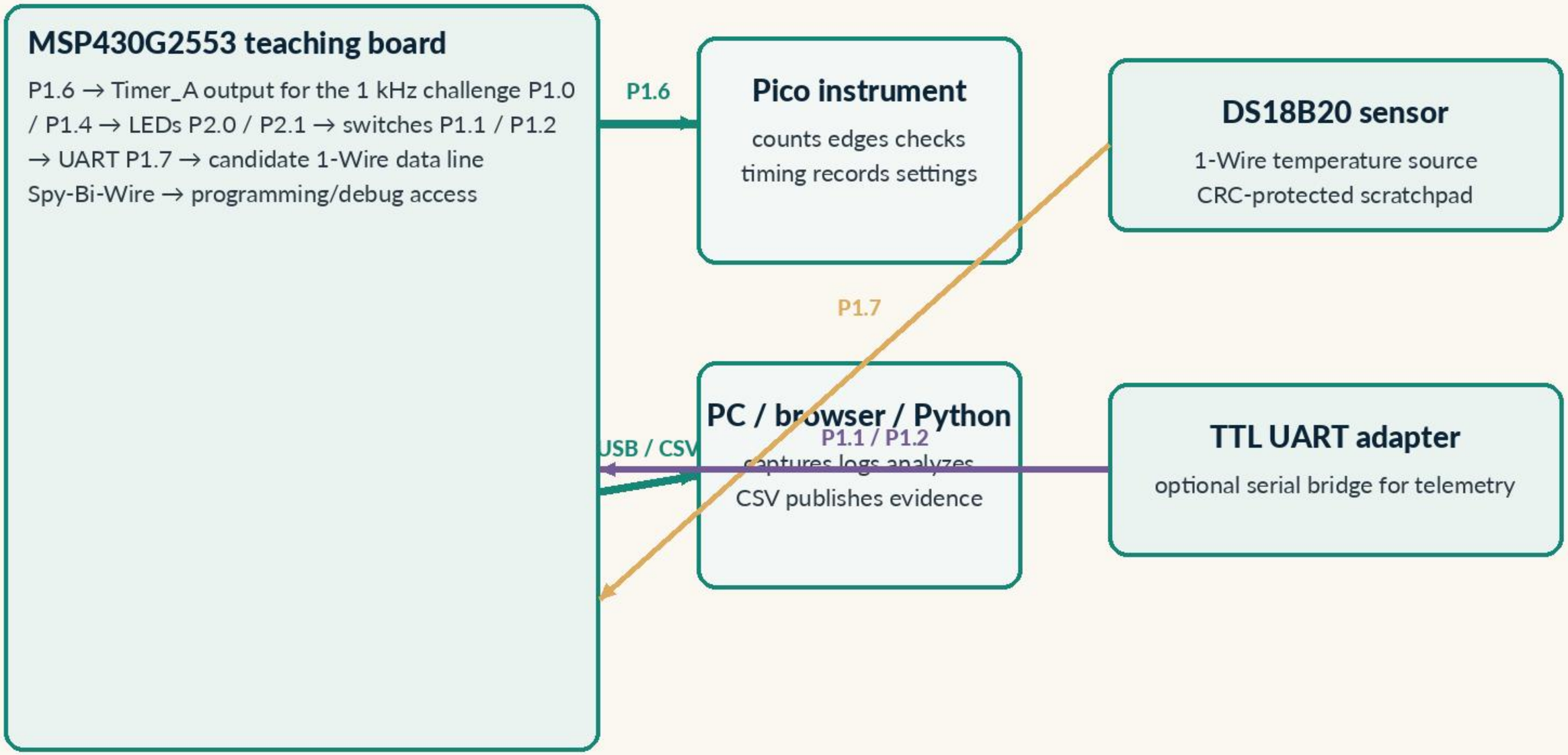
Read this plate while tracing assembly

Mode	PDP-11 example	MSP430 example
Register	`MOV R4,R5`	`MOV.W R4,R5`
Immediate	`#10,R4`	`#10,R4`
Indexed	`4(R5),R6`	`4(R5),R6`
Register indirect	`(R5),R6`	`@R5,R6`
Autoincrement source	`(R5)+,R6`	`@R5+,R6`
Destination limits	`MOV R4,(R5)+`	`not generally symmetric`

FIG 5

MSP430 laboratory bench map

Where signals and responsibilities live



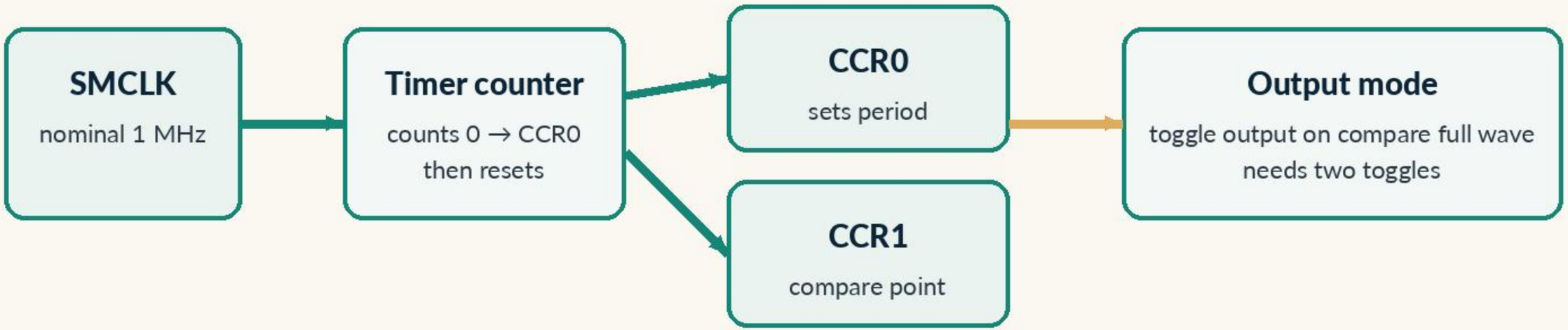
Design rule

- The controller, instrument, and publication path have different jobs. The Pico does not replace the MSP430, and the host PC does not prove timing just by showing a formatted number.

FIG 6

Timer_A concept map

From SMCLK to a square wave



Worked example

If SMCLK = 1,000,000 Hz and CCR0 = 499, the counter visits 500 states per interval. The output toggles once per interval, so the time between like edges is two intervals. Nominal frequency = 1,000,000 / [2 × (499 + 1)] = 1,000 Hz.

FIG 7

Active-low GPIO truth table

One bit can mean different things electrically and logically

Physical condition	Pin voltage	P2IN bit	Program test	Meaning
Switch open	HIGH	1	(P2IN & BIT0) != 0	not pressed
Switch pressed to GND	LOW	0	(P2IN & BIT0) == 0	pressed
Wire broken / floating	unknown	unknown	indeterminate	fault candidate
Short to ground	LOW always	0 always	appears pressed	sensor fault

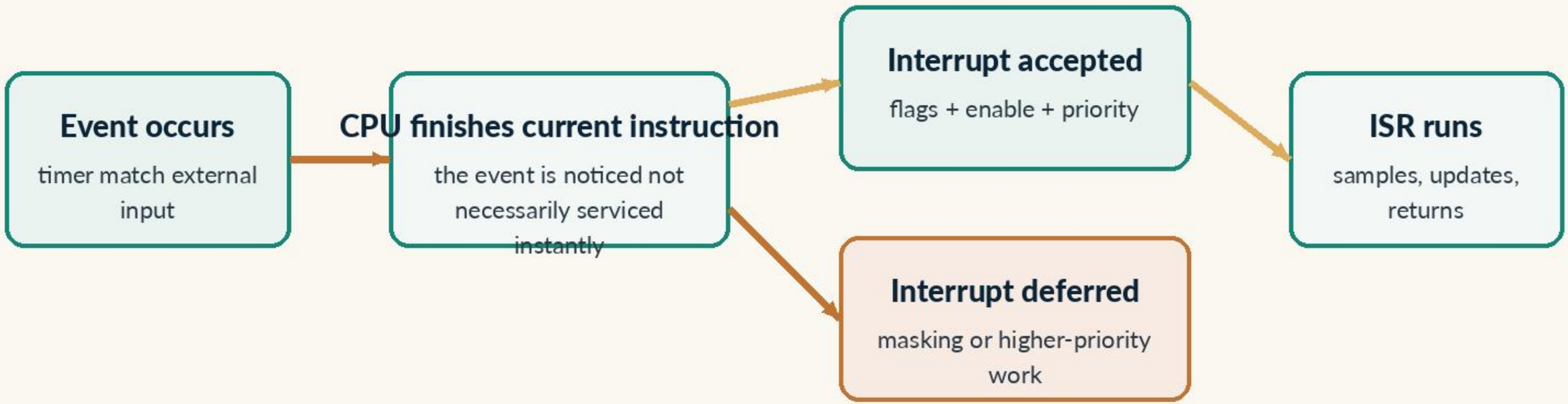
Teaching point

- A Boolean variable such as pressed is an interpretation layered on top of a measured voltage and a sampled register bit.
- Industrial digital inputs often need the same discipline: electrical polarity, semantic meaning, and fault behavior must be documented separately.

FIG 8

Interrupt flow and latency

Why deadlines are about response, not average speed



What students should conclude

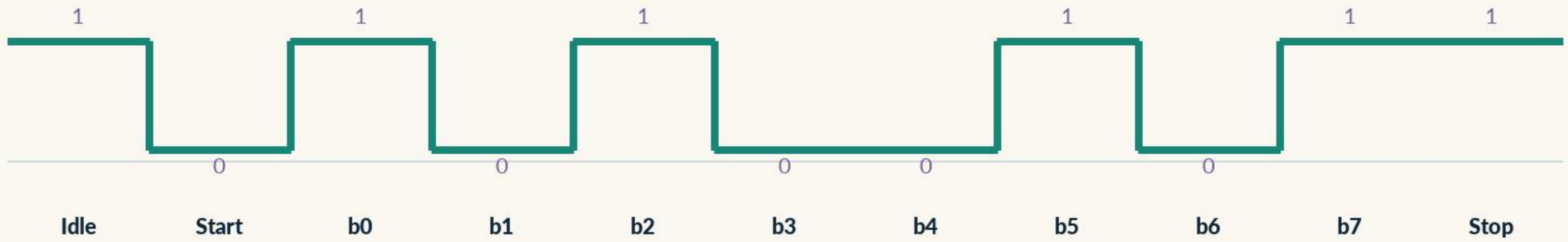
Latency = event to service start. Jitter = variability in timing from one event to the next. A hardware timer output can have low jitter even while an interrupt-driven software output looks noisy. Therefore “it averages correctly” does not answer whether a protocol slot or control deadline was met.

FIG 9

UART frame anatomy

A serial byte as a timed waveform

Line idle is HIGH. The transmitter pulls LOW for the start bit, then sends data least-significant bit first.



Interpretation

One framed byte is meaningless without clock assumptions and shared conventions. Correct characters depend on the chosen baud rate, voltage compatibility, common ground, and a receiver that samples near the center of each bit cell.

FIG 10

1-Wire reset and presence timing

Ownership of the line matters as much as the bit values

The master drives low for reset, then releases. The pull-up restores HIGH. The sensor may then pull LOW to signal presence.



idle reset low release presence idle

Master drives low

Electrical action: pull bus low for a valid reset interval.

Master releases line

The pull-up, not the master, restores HIGH.

Sensor may drive presence low

A low pulse in the expected window indicates a responding device.

FIG 11

1-Wire write and read slots

Short low pulses and sample windows are not the same event

Write-1 slot

Master pulls low briefly, then releases early so the line is HIGH for most of the slot. The receiver interprets the slot as a logical 1 because the line is not held low for long.



Write-0 slot

Master holds low for much longer within the slot. The low interval itself carries the information. Timing margins matter more than average frequency.



Read slot

Master initiates the slot with a short low, then releases. The sensor may pull low. The master must sample during the valid window, not merely observe that the line changed at some point.

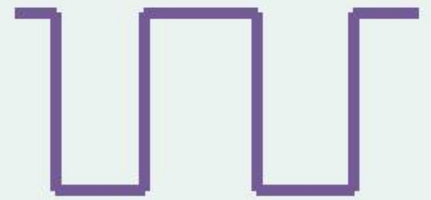


FIG 12

DS18B20 scratchpad map

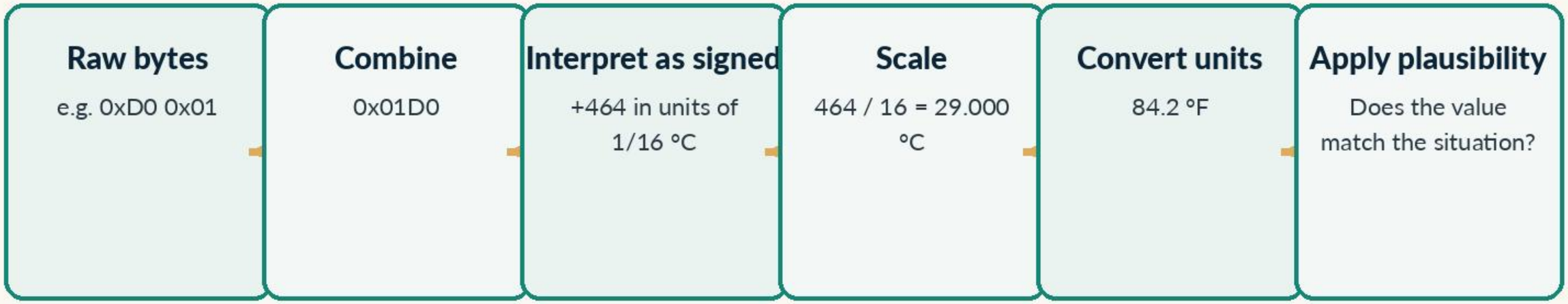
Nine bytes and what they mean

Byte	Common name	Meaning	Teaching note
0	Temp LSB	least-significant byte of temperature	Combine bytes 0 and 1 as a signed 16-bit value.
1	Temp MSB	most-significant byte of temperature	Negative values require two's-complement interpretation.
2	TH	alarm high register	Configuration context, not current temperature.
3	TL	alarm low register	Configuration context, not current temperature.
4	Config	resolution bits among other fields	Resolution affects meaningful low bits and conversion time.
5	Reserved	implementation-defined / reserved	Do not invent a meaning.
6	Reserved	implementation-defined / reserved	Read and preserve as documented.
7	Reserved	implementation-defined / reserved	Included in the CRC calculation.
8	CRC	integrity byte	Valid CRC means the bytes arrived consistently, not that the temperature is accurate.

FIG 13

From raw bytes to engineering units

The decoding path should be explicit



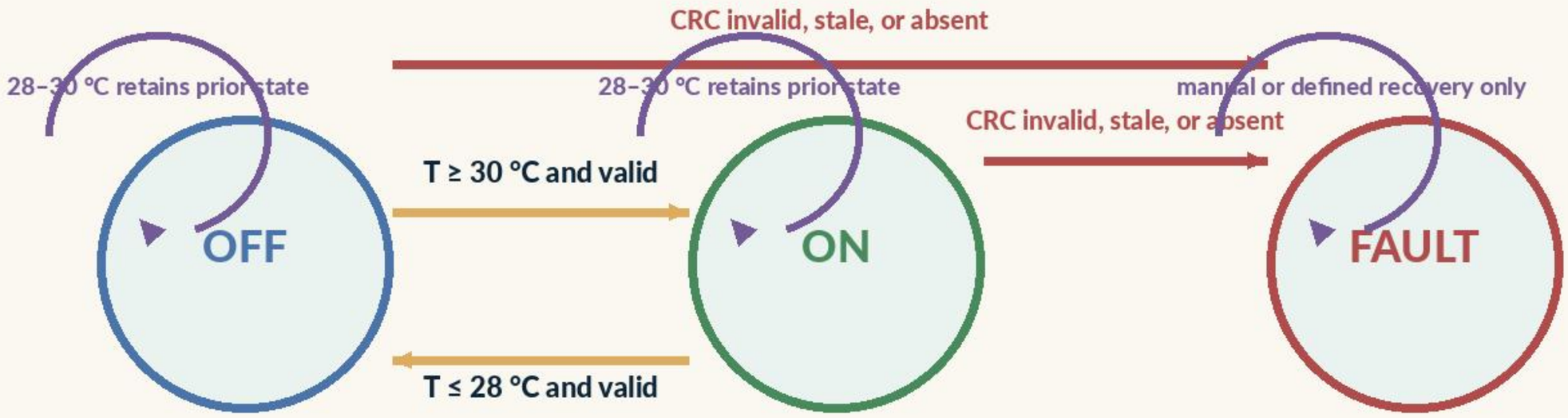
Caution

CRC-valid data can still describe the wrong thing: a stale conversion, a poorly placed sensor, or the thermal behavior of the sensor package rather than the process. Arithmetic correctness and instrument truth are related but distinct achievements.

FIG 14

Temperature supervisor state machine

A small but authentic control design



Curricular value

This one plate lets students discuss memory, thresholds, fault policy, and why a host-side unit test can validate logic without proving the full electrical controller.

FIG 15

Capture → Analyze → Publish

The evidence model of the course



Rules of use

- Capture is about what happened, not what the model predicted should happen.
- Analyze compares observation with theory and identifies what remains uncertain.
- Publish must preserve enough setup detail that another person can challenge or reproduce the result.

FIG 16

Evidence ladder

What different kinds of artifacts can and cannot establish

How to use this figure

Students often mistake a successful compile or a plausible display for a fully established result. This ladder helps them name what level of support they actually have.

Characterized instrument result — Calibration, uncertainty, and traceability are discussed.

Replicated result — A second controlled run agrees within limits.

Recorded measurement — Raw data and settings are preserved.

Observed display — The instrument shows a value.

Source artifact — A listing exists, but has not been run here.

Claim only — “The code should make 1 kHz.”

FIG 17

Suggested figure placement by chapter

A quick guide for using the atlas in the textbook

Chapters	Recommended figures	Purpose
Ch. 1–3	FIG 1, FIG 2, FIG 16	industrial computing, architecture, evidence
Ch. 4	FIG 3, FIG 4	PDP-11 ↔ MSP430 comparison
Ch. 5	FIG 5, FIG 6	bench map and Timer_A concept
Ch. 6–8	FIG 7, FIG 8, FIG 9	GPIO, interrupts, UART
Ch. 9–10	FIG 10, FIG 11, FIG 12, FIG 13	1-Wire and DS18B20
Ch. 11–14	FIG 14, FIG 15, FIG 16	state machine, CAP, publication

Editorial note

These figures are designed to teach, not merely decorate. The best use is to pair each plate with a chapter question or laboratory prediction task.